

ESEQUIEL LINARES

West Lafayette, IN | 312-241-4367 | esequellinares06@gmail.com
www.linkedin.com/in/esequiel-linares | www.zekeprojects.com | www.github.com/Zekepeke

EDUCATION

Purdue University

West Lafayette, IN

Bachelor of Science in Computer Science

May 2028

- **Relevant Coursework:** Data Structures & Algorithms, Computer Architecture, Linear Algebra, Multivariate Calculus, Discrete Math, Intro Artificial Intelligence, Systems Programming

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, C, JavaScript, TypeScript, SQL, Swift, Bash

Frameworks & Libraries: FastAPI, Flask, Spring Boot, React, Next.js, Node.js, NumPy, Pandas, Mediapipe, JUnit, pytest

AI/ML: PyTorch, Hugging Face, llama.cpp, faster-whisper, Claude API, Gemini API, RAG, Prompt Engineering

Tools & Platforms: Git, Docker, Databricks, Delta Lake, PySpark, Jenkins, Bitbucket, AWS (EC2), GCP, Supabase, Modal, Linux/Unix, PostgreSQL

EXPERIENCE

Software Engineering Intern

June 2026 – August 2026

JPMorgan Chase & Co.

Chicago, IL

- Design and build the ingestion layer of Hermes, an internal document processing platform converting invoices and contracts (PDF, PNG) into schema-validated Delta tables, applying Strategy, Factory, and Adapter patterns so new file formats and parsers drop in without pipeline code changes.
- Streamline end-to-end processing for a 100-document batch from ~2 hours to 11 minutes by profiling the bottleneck, parallelizing extraction across 4–8 workers, and consolidating per-record Delta writes into 300-record merge operations.
- Implement validation gates for the LLM mapping job, layering deterministic field checks with an LLM verification pass and routing malformed records to a Dead Letter Queue, with JSON schemas and guardrail prompts loaded from Delta tables rather than hardcoded.
- Present the platform to senior firm leadership, securing handoff to a partner team for production adoption.
- Coach two high school interns building the OCR extraction path (RapidOCR, PNG support) into the shared extraction interface.

Undergraduate Researcher – Multimodal Generative AI

September 2025 – May 2026

Purdue University

West Lafayette, IN

- Extended PCMDM, a past-conditioned diffusion model for coherent long-form 3D human motion generation, training a variant on a 10K+ annotation duet dance dataset that improved R-Precision Top-3 by 60% relative to the provided baseline.
- Built PyTorch training and evaluation pipelines for Text-to-Duet and Dance Accompaniment, implementing an R-Precision harness that ranks ground-truth captions against mismatched candidates in a shared text-motion embedding space.

PROJECTS

Reachy Mini | *Raspberry Pi 5, RUBIK Pi 3 (QCS6490), llama.cpp, faster-whisper, Gemini Vision, ElevenLabs, Python*

- Led the software architecture for a low-latency voice-and-vision assistant across a Raspberry Pi 5 orchestrator and a Qualcomm QCS6490 AI coprocessor, linked by custom HTTP endpoints for remote speech-to-text and LLM inference.
- Benchmarked on-device LLMs and selected a quantized 3B model (5.2 tok/s, 8K context) over an 8B variant, preserving agentic tool-use.
- Diagnosed a 0.04x-realtime on-device TTS path on the Adreno 643 GPU and re-architected it to sentence-streaming cloud TTS, cutting time-to-first-audio from ~60s to under 2s for near-real-time conversation.

Price is Right Glasses | *FastAPI, Claude/Gemini Vision, Supabase, SwiftUI, ElevenLabs*

- Built a multimodal price-check assistant for Meta Ray-Ban glasses that identifies items by vision, queries live pricing APIs (eBay, Discogs, TCG), and speaks a buy verdict aloud.
- Engineered a provider-agnostic vision layer with swappable Claude and Gemini models behind one prompt and response schema, served by FastAPI.

machine(learn) | *Modal, Next.js, TypeScript, Supabase, Pydantic*

- Built an autonomous ML agent swarm on Modal GPU infrastructure, with a four-phase async orchestrator using Pydantic contracts for typed agent handoffs.
- Streamed live run state to a Next.js dashboard over Supabase realtime, with authentication and per-run dataset uploads.

LEADERSHIP & PROFESSIONAL DEVELOPMENT

Career Prep Fellow

January 2026 – Present

Management Leadership for Tomorrow

- Selected for an 18-month program accelerating emerging technical leaders through coaching, technical assessments, and project work.
- Engage with partner organizations (e.g., LinkedIn, Bloomberg, Deloitte) to gain industry exposure and insight into tech-talent pathways.

Backend Developer – Code for Good Hackathon

November 2025

JPMorgan Chase & Co.

Columbus, OH

- Developed 10+ Flask RESTful endpoints with SQLAlchemy ORM and an automated CSV ingestion pipeline converting raw data into structured records for real-time dashboard updates.
- Authored 56 commits (47% of repository), leading backend architecture and deploying to AWS EC2 to achieve top-3 placement in 24 hours.